

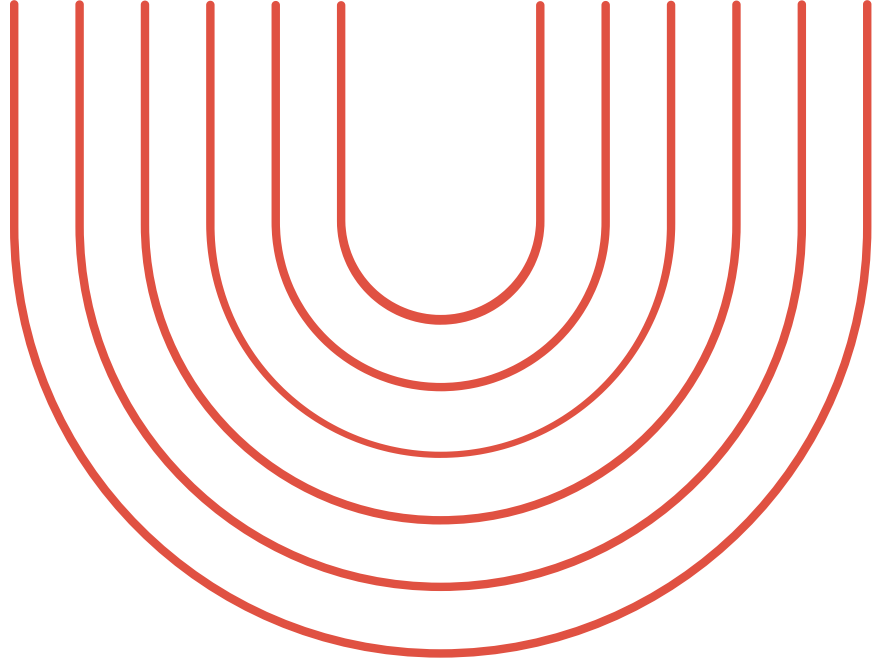


**ATLIQ
HARDWARE**

CONSUMER GOODS AD-HOC INSIGHTS

SQL PROJECT CHALLENGE

- Vishwash Chuahan



ABOUT ATLIQ HARDWARE

Company

Atliq Hardwares (Imaginary company) - One of the leading computer hardware producers in India.

Background

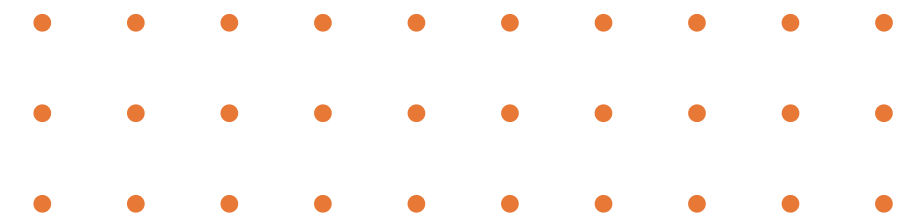
The management noticed that they do not get enough insights to make quick and smart data-informed decisions.

Problem

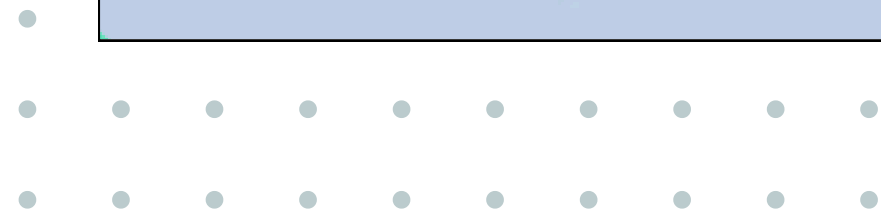
There are 10 ad-hoc requests for which the company needs insights.

Approach

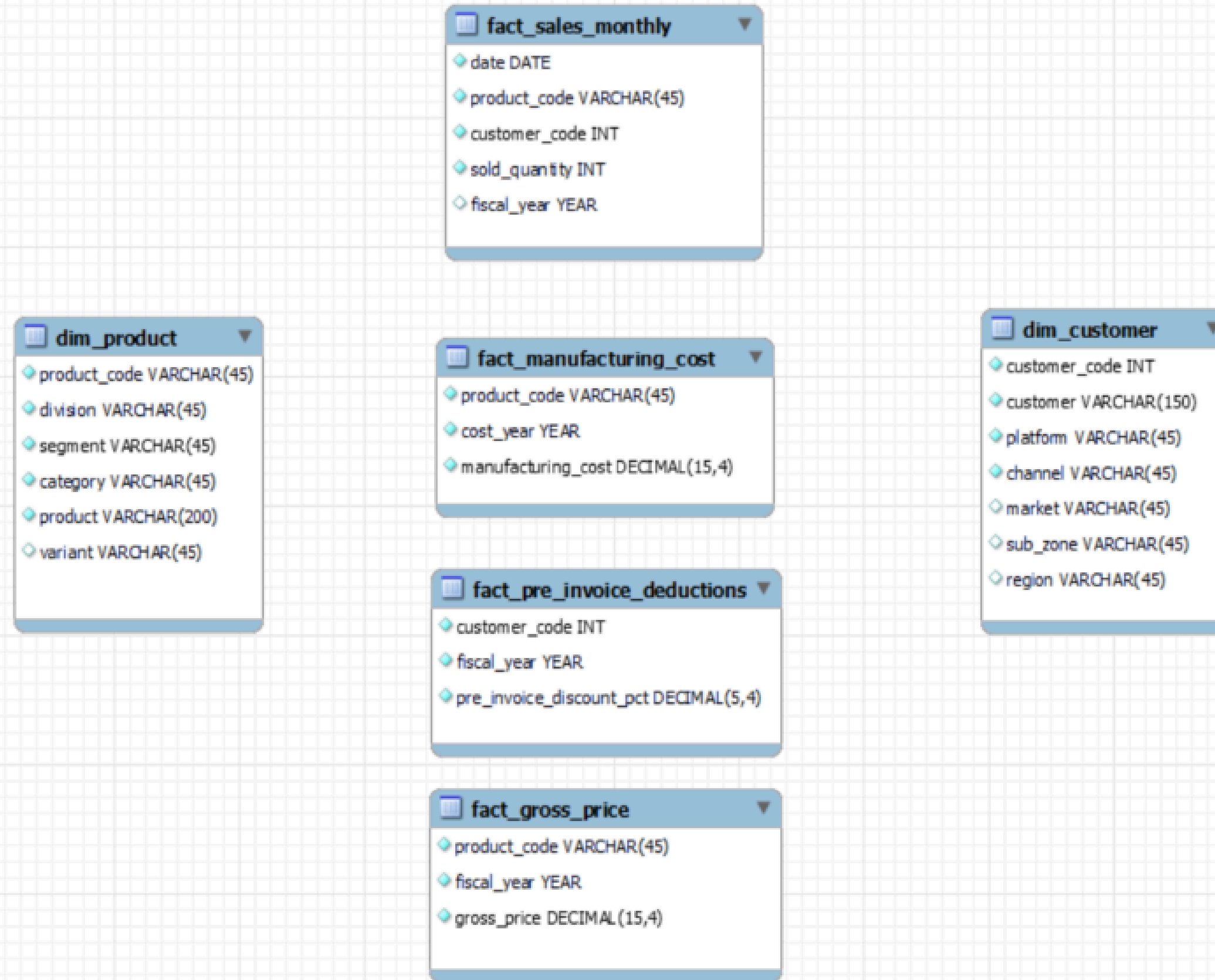
Run a SQL query to answer these requests. Convert it into visualizations and present the Insights to the top-level management.



A stylized illustration of a blue sky with white clouds and a green hill in the foreground. The sky is a solid light blue, and the clouds are simple white shapes. The hill is a solid green shape at the bottom left. The background is a solid light blue.

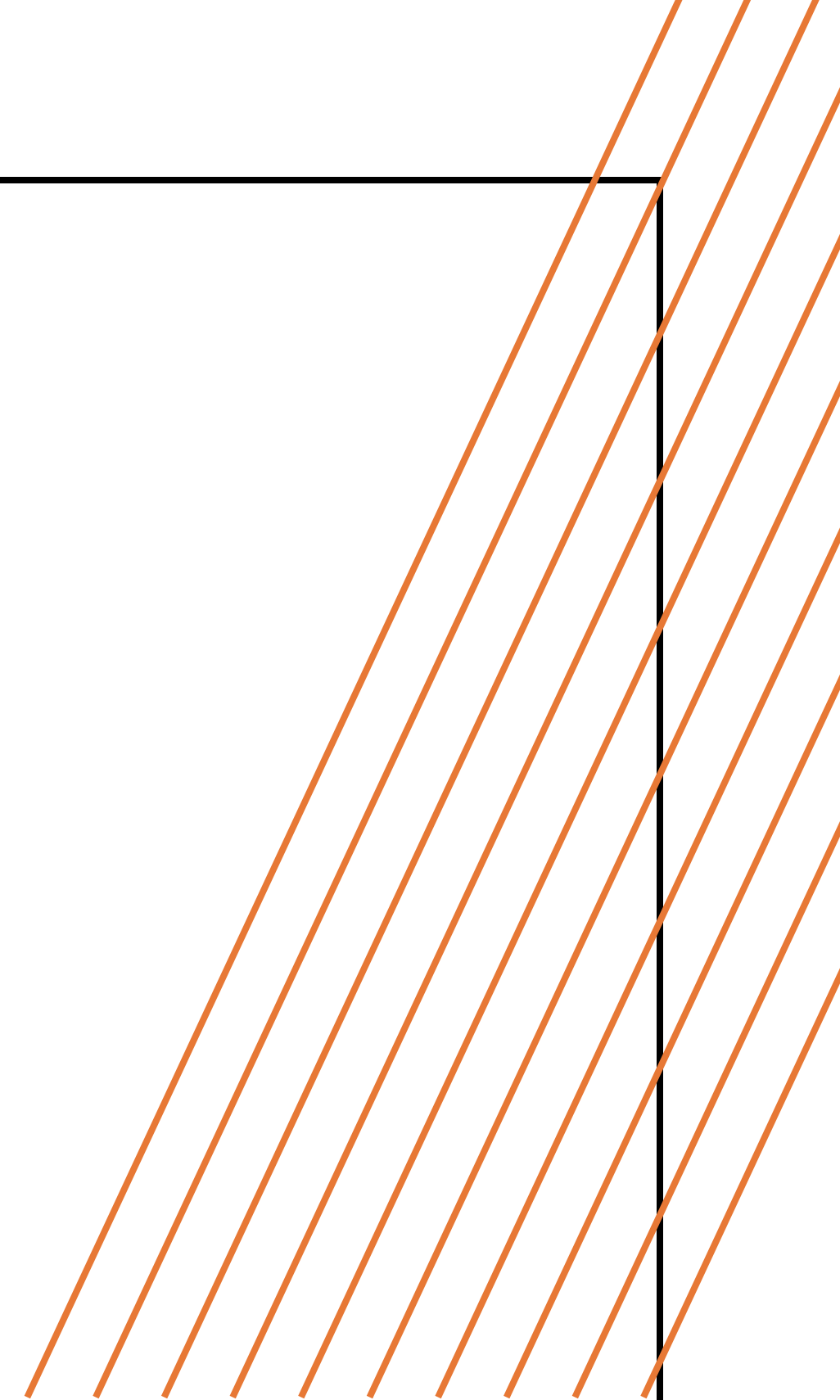
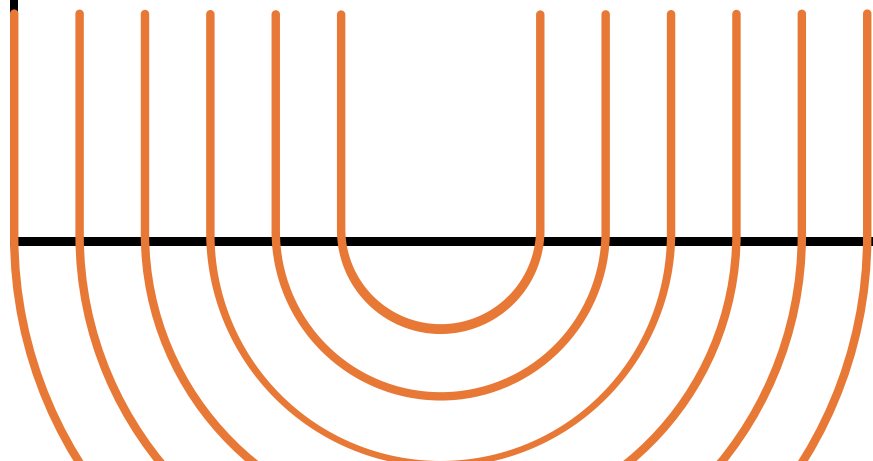


DATASETS



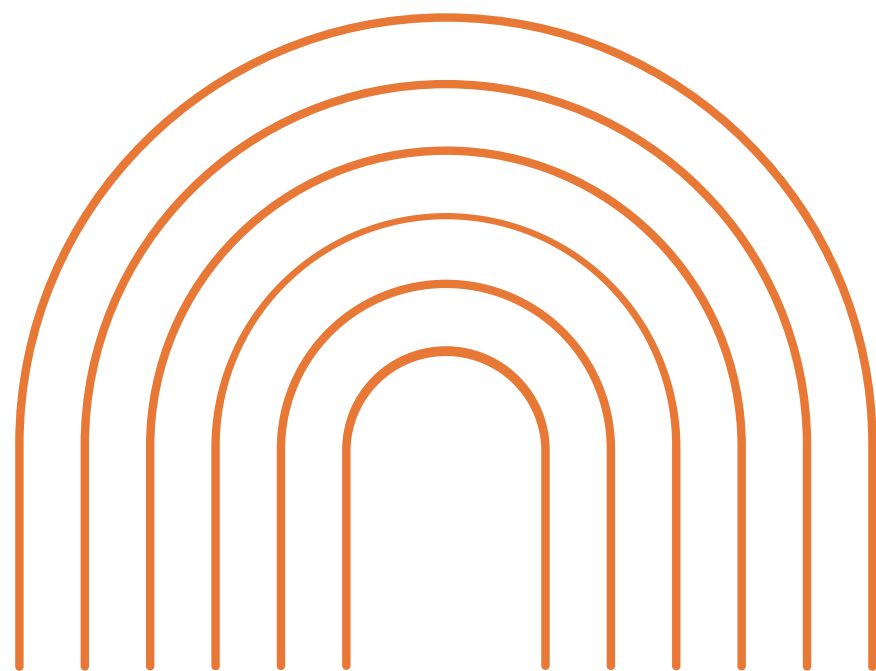
Input data consists of sales data for FY 2020 and FY 2021, along with different other dimension tables like customer details, product details, etc.

**AD-HOC
REQUESTS,
QUERIED RESULTS,
INSIGHTS AND
VISUALIZATION**



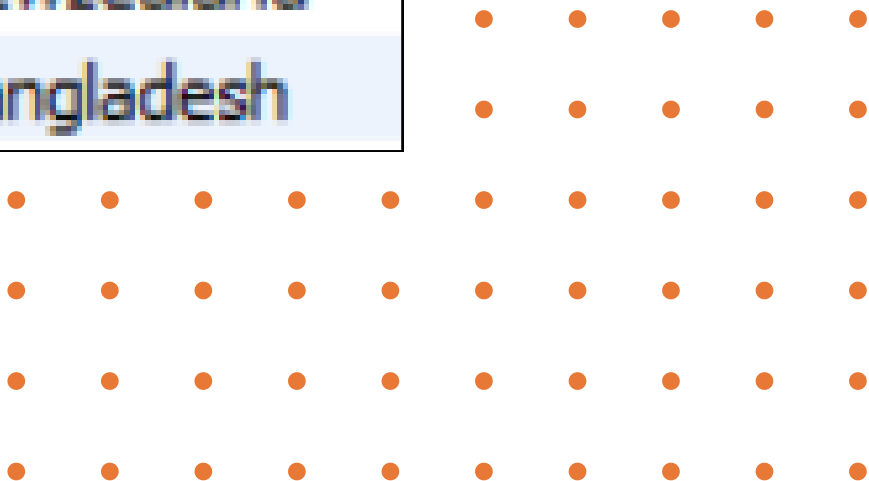
REQUEST 01.

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.



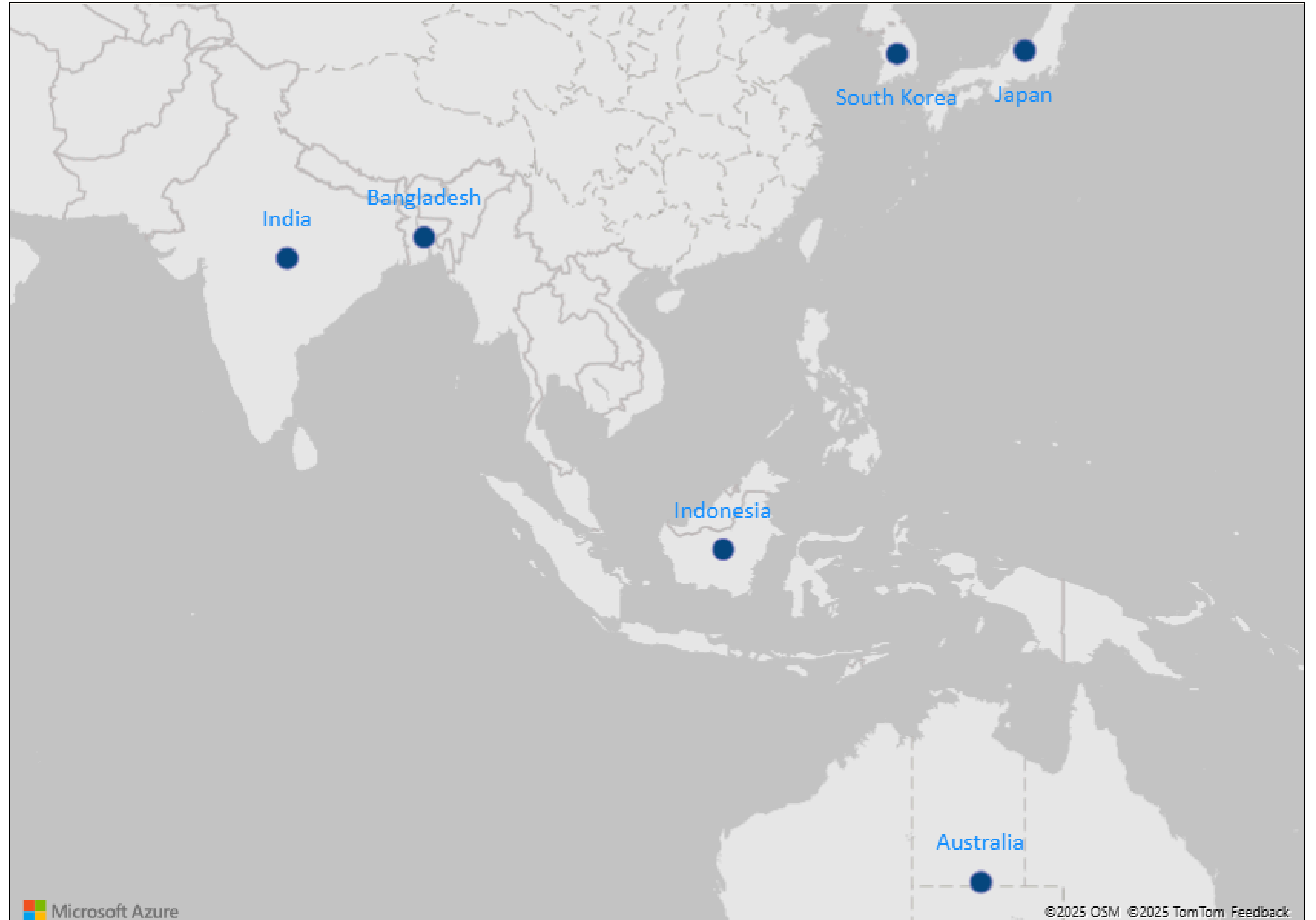
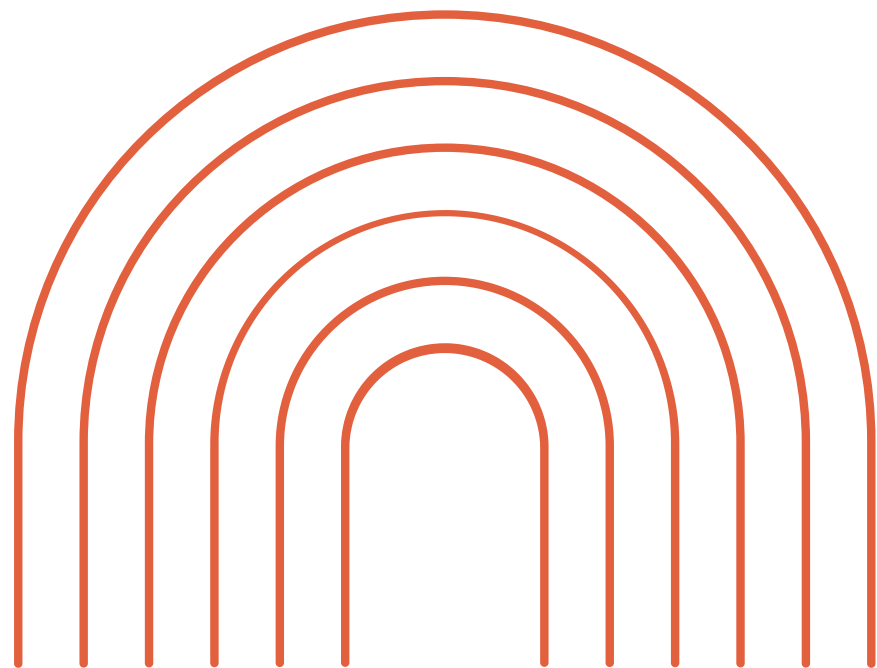
OUTPUT

	market
▶	India
	Indonesia
	Japan
	Philippines
	South Korea
	Australia
	Newzealand
	Bangladesh



INSIGHTS

AtliQ Exclusive stores
operate in **8 countries**
within the **APAC**
region.



REQUEST 02.

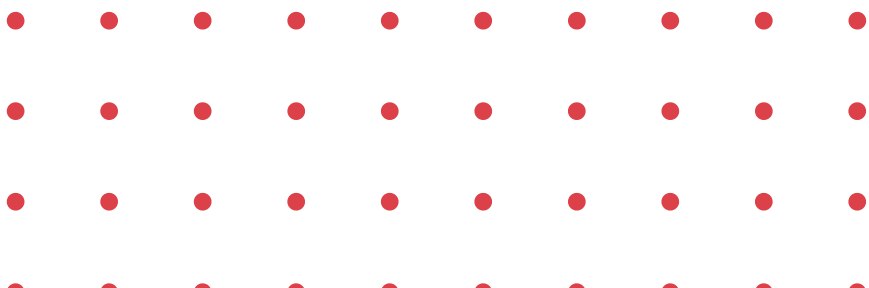
What is the percentage of unique product increase in 2021 vs. 2020?

The final output contains these fields:

unique_products_2020
unique_products_2021
percentage_chg

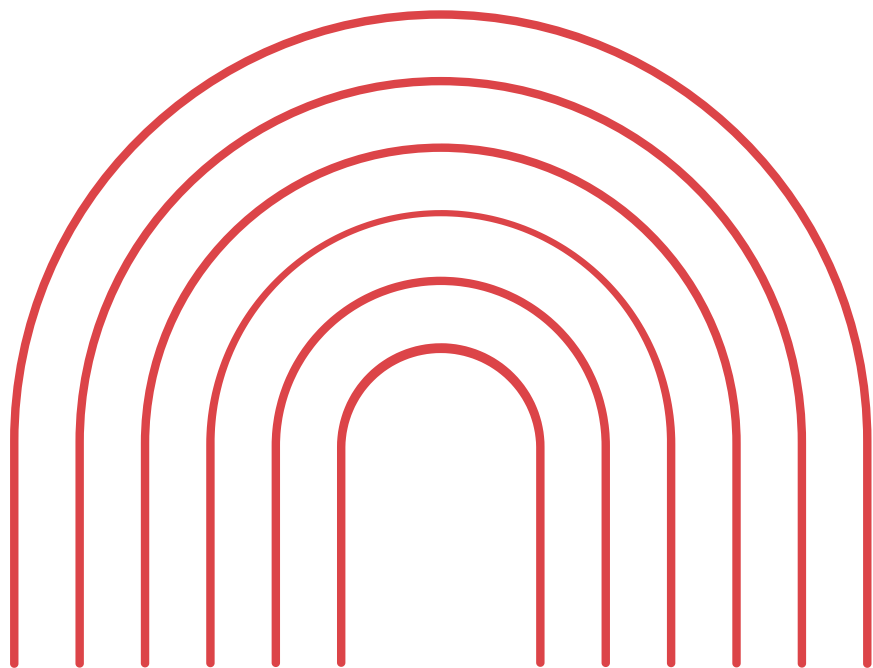
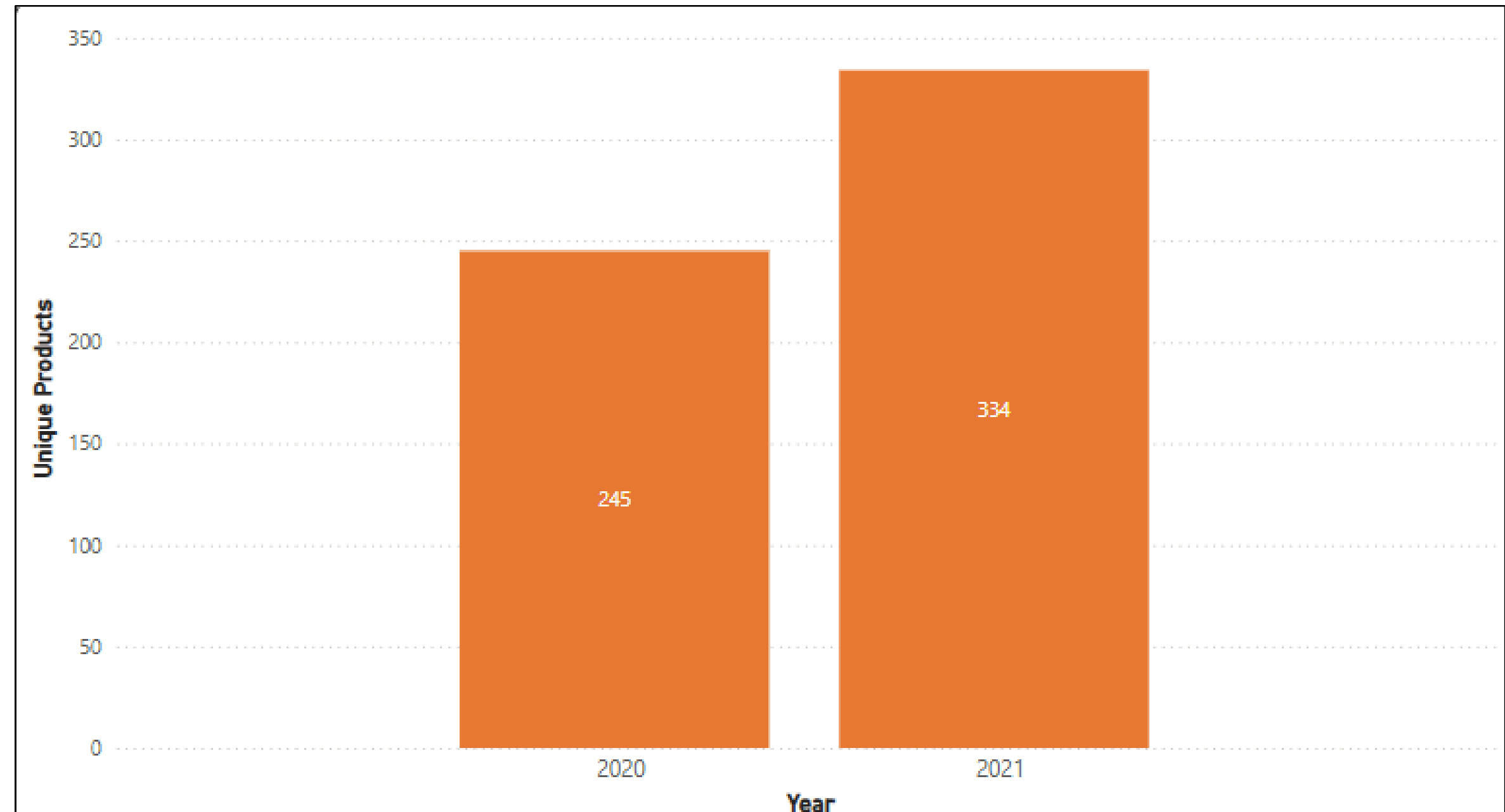
OUTPUT

	unique_products_2020	unique_products_2020	percentage_chg
▶	245	245	36.33%



INSIGHTS

In FY 2020, total number of products was 245, which was increased by 36% in FY 2021.





REQUEST 03.

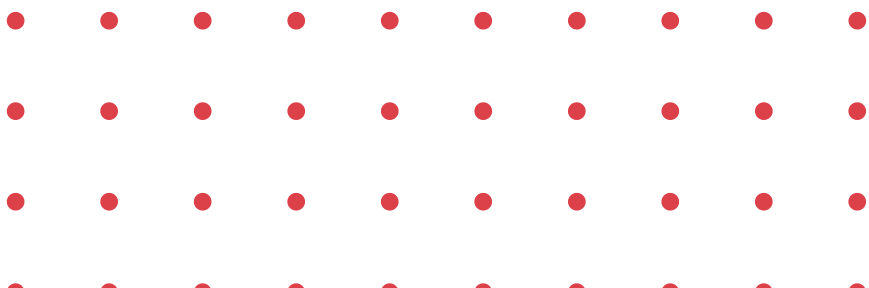
Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

The final output contains 2 fields:

Segment
product_count

OUTPUT

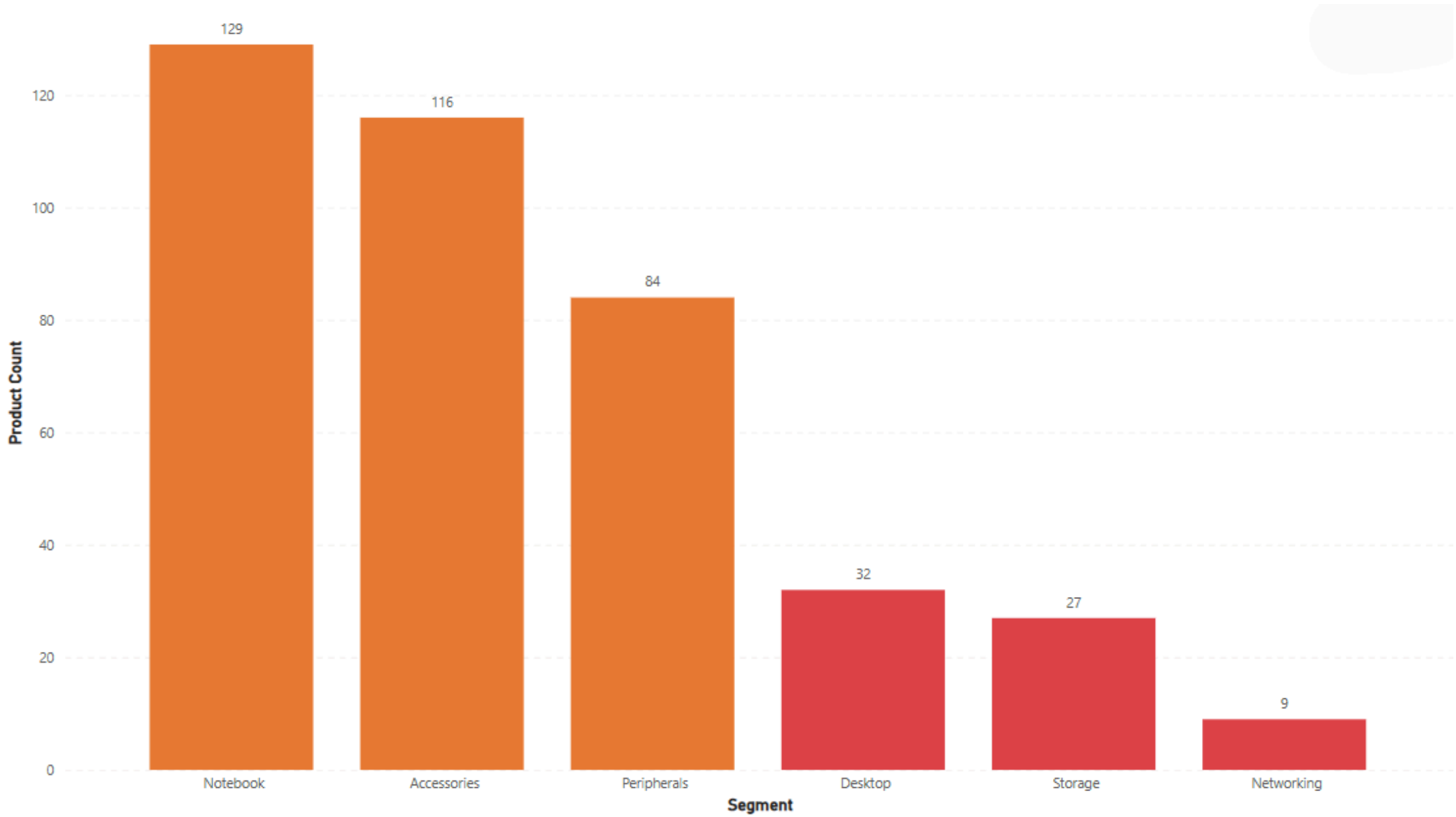
	segment	product_count
▶	Notebook	129
	Accessories	116
	Peripherals	84
	Desktop	32
	Storage	27
	Networking	9



INSIGHTS

Atliq provide a wide range of products under the segment Notebook, Peripherals, and Accessories, with an average of 110 products in each segment.

However, the company still need to diversify its product offering in the Desktop, Networking, and Storage segments, where there are just an average of only 23 products per segment.



REQUEST 04.

Follow-up:
Which segment had the
most increase in unique products in
2021 vs 2020?

The final output contains these fields:

segment
product_count_2020
product_count_2021
difference

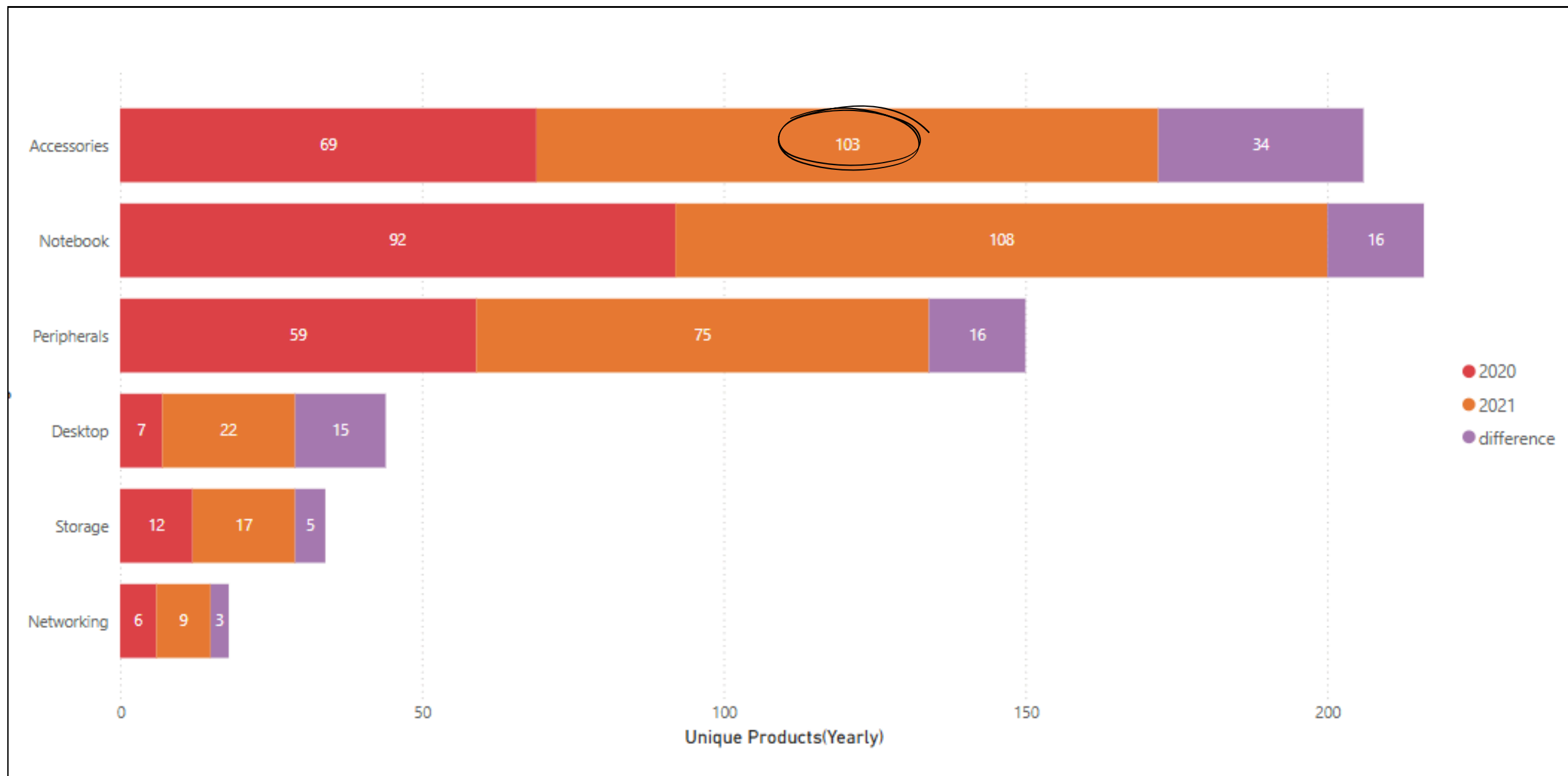
OUTPUT

	segment	product_count_2020	product_count_2021	difference
▶	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3



INSIGHTS

With the introduction of 34 new products, Accessories segment has the highest increase in number of unique products during FY 2020 - 2021





REQUEST 05.

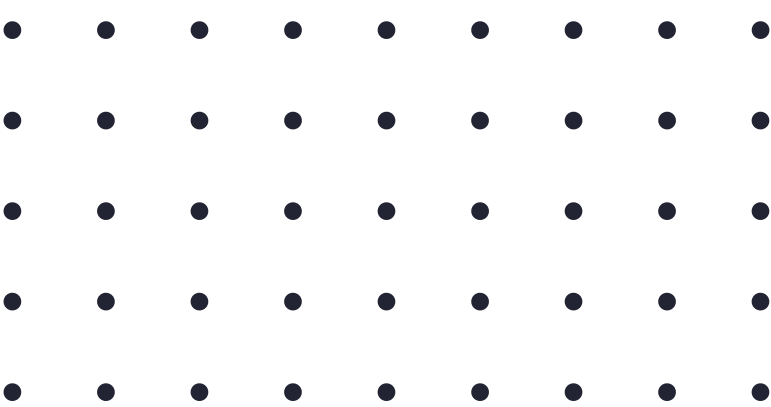
Get the products that have the highest and lowest manufacturing costs.

The final output should contain these fields:

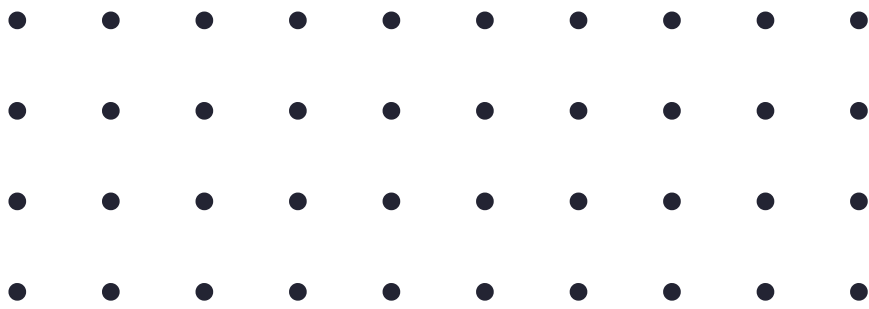
product_code
product
manufacturing_cost

OUTPUT

	product_code	product	manufacturing_cost
▶	A6120110206	AQ HOME Allin1 Gen 2	\$240.54
	A2118150101	AQ Master wired x1 Ms	\$0.89



INSIGHTS



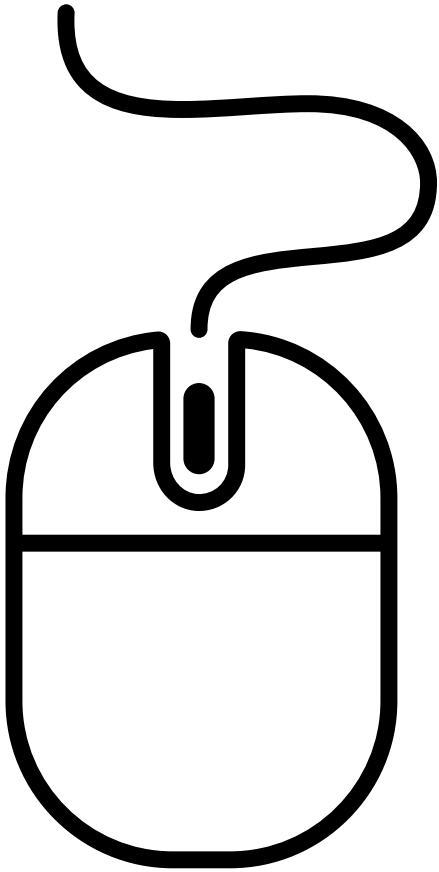
HIGHEST MANUFACTURING
COST



AQ HOME Allin1 Gen 2 (Plus 3)
Category: Personal Desktop

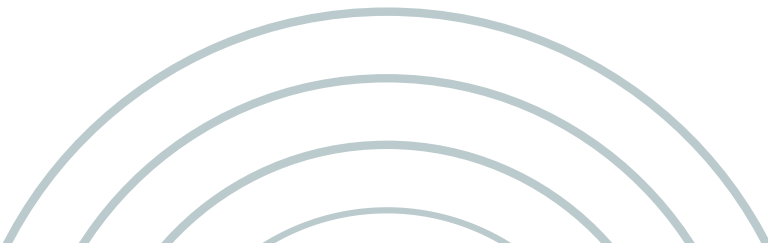
\$240.54

LOWEST MANUFACTURING
COST



AQ Master wired x1 Ms (Standard 1)
Category: Mouse

\$0.98



REQUEST 06.

Generate a report which contains the top 5 customers who received an average high pre_invoice discount_pct for the fiscal year 2021 and in the Indian market.

The final output contains these fields:

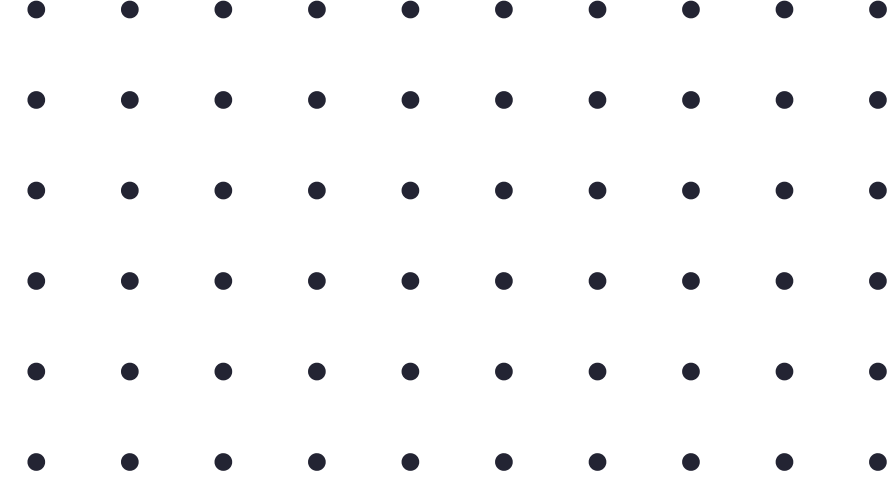
customer_code
customer
average_discount_percentage

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OUTPUT

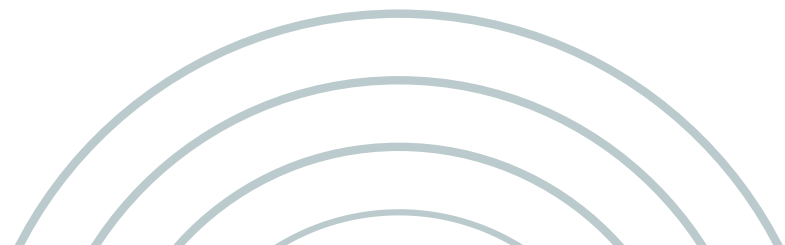
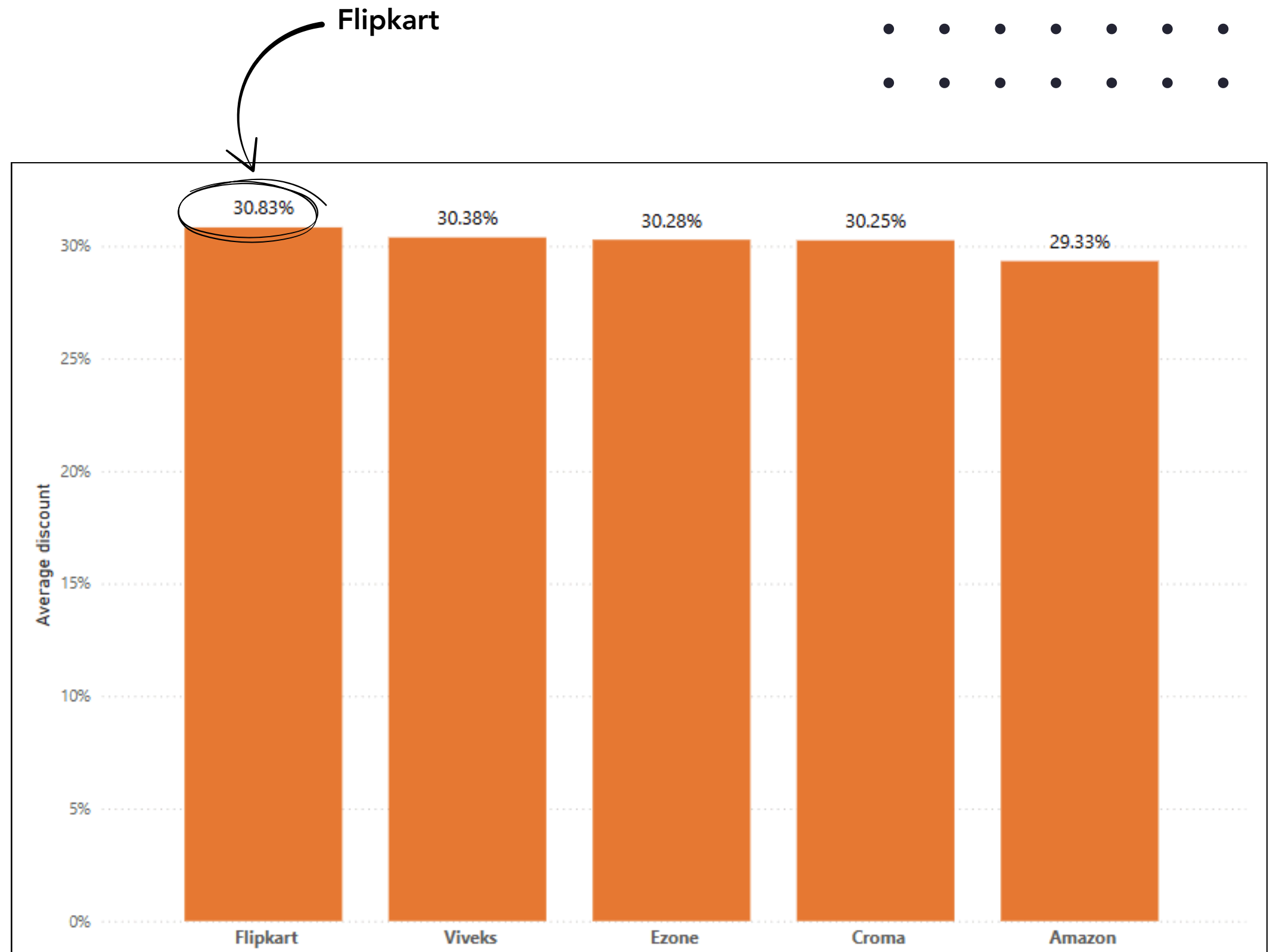
	customer_code	customer	average_discount_percentage
▶	90002009	Flipkart	30.83%
	90002006	Viveks	30.38%
	90002003	Ezone	30.28%
	90002002	Croma	30.25%
	90002016	Amazon	29.33%

INSIGHTS



Top 5 Customers have a collective average around 30.21%

Flipkart has received the highest pre invoice discount percent i.e., 30.83%.



REQUEST 07.

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.

The final report contains these columns:

Month
Year
Gross sales Amount

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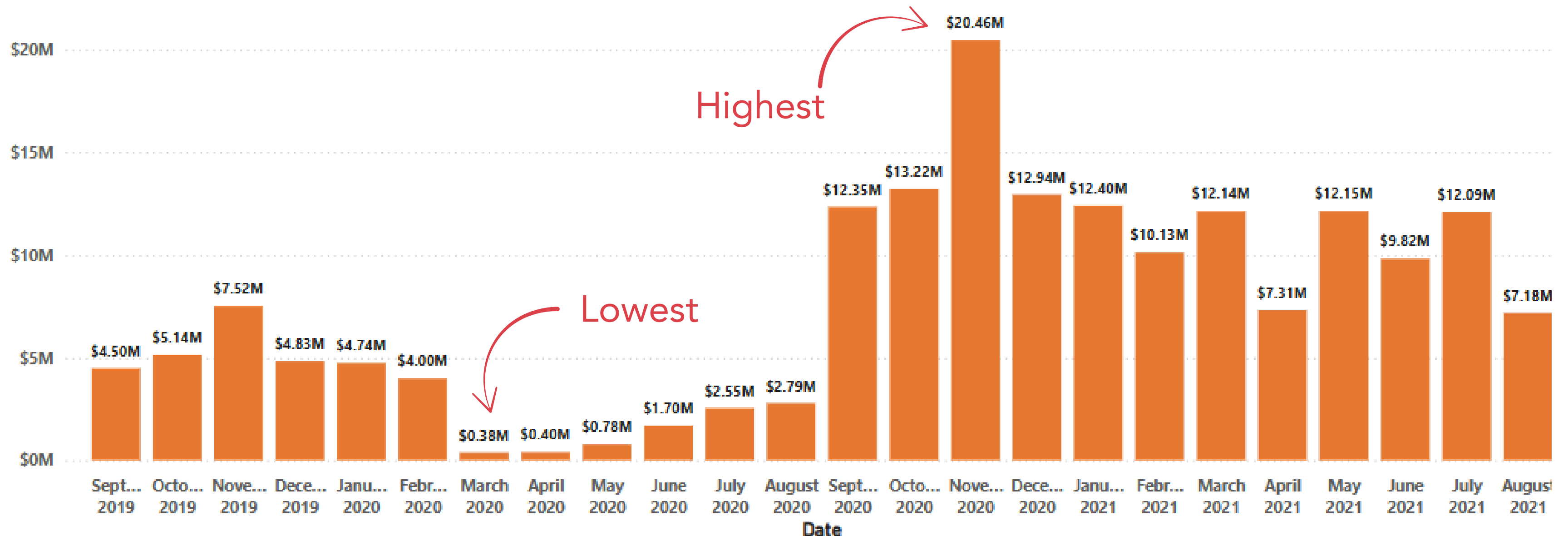
OUTPUT

	month_name	year_	gross_sales_amount_millions
	October	2019	\$5.14
	November	2019	\$7.52
	December	2019	\$4.83
	January	2020	\$4.74
	February	2020	\$4.00
	March	2020	\$0.38
	April	2020	\$0.40
	May	2020	\$0.78
	June	2020	\$1.70
	July	2020	\$2.55
	August	2020	\$2.79
	September	2020	\$12.35
	October	2020	\$13.22
	November	2020	\$20.46
	December	2020	\$12.94
	January	2021	\$12.40

INSIGHTS

For Atliq Exclusive, **November 2020** marked the **highest sales**, and **March 2020** marked the **lowest sales**.

It's very evident that the lower sales between March and August are because of COVID-19. However, it's a very good sign that the sales increased quickly after August and reached the highest level since the last two years in November.



REQUEST 08.

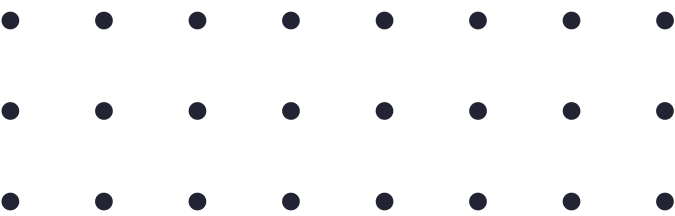
In which quarter of 2020, got the maximum total_sold_quantity?

The final output contains these fields sorted by the

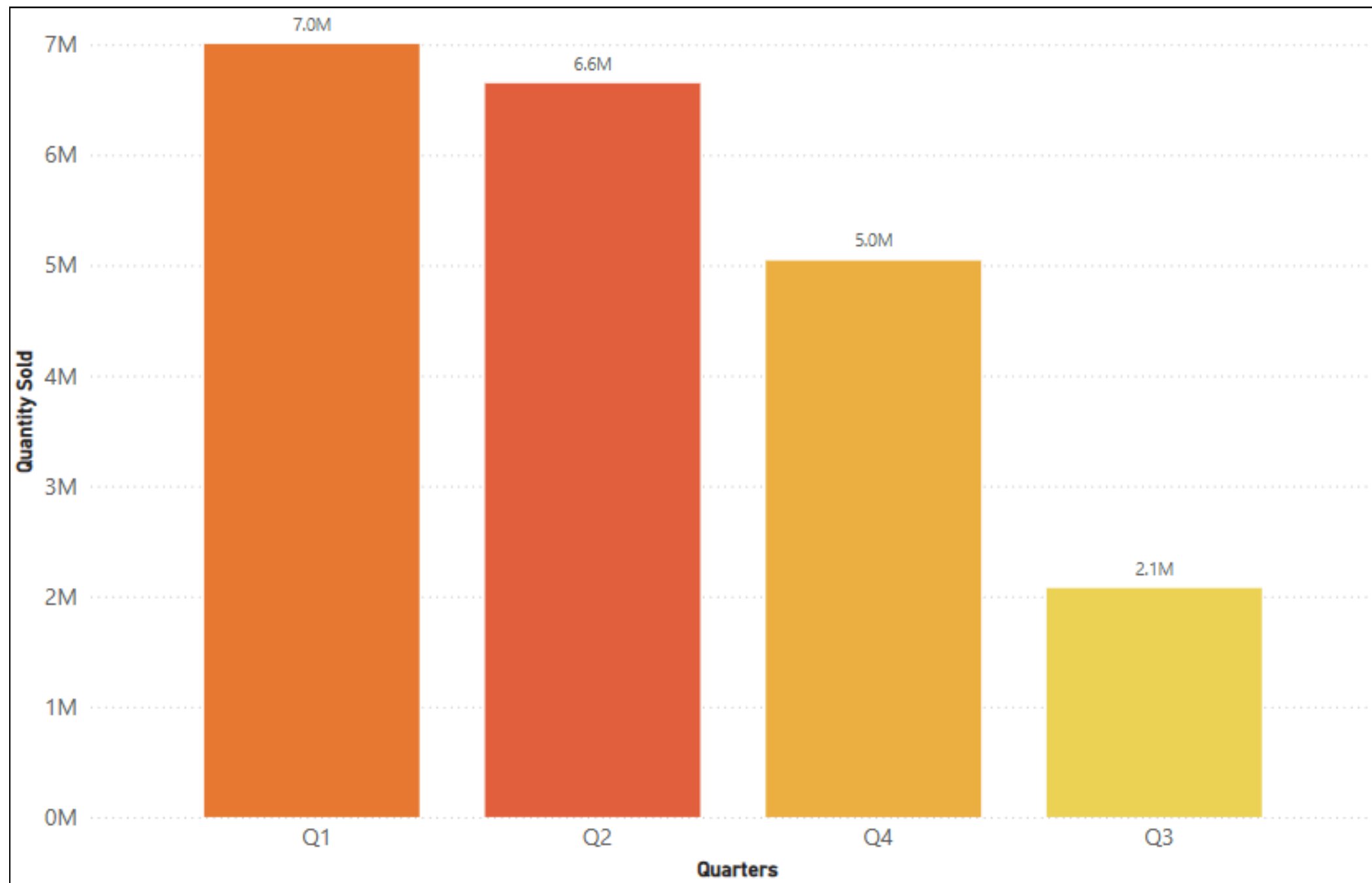
total_sold_quantity:
Quarter
total_sold_quantity

OUTPUT

	quarters	total_quantity_sold
▶	Q1	7005619
	Q2	6649642
	Q4	5042541
	Q3	2075087

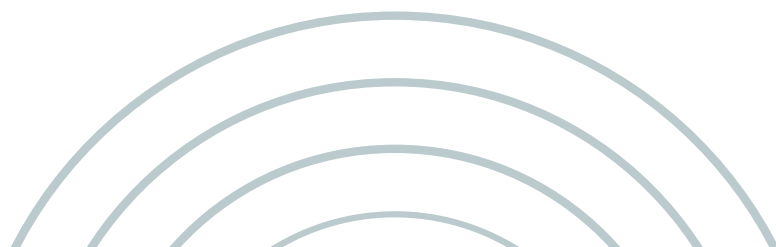


INSIGHTS



Q1 (September-November) had the maximum quantity sold for FY 2020 standing at **7 millions**.

The sold quantity decreased to 2.1 million in quarter 3 of FY 2020, which was actually March, April, and May when COVID-19 was at its peak.



REQUEST 09.

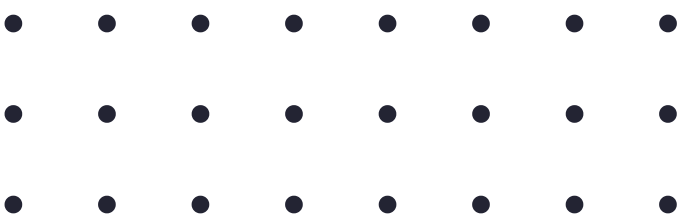
Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

The final output contains these fields:

channel
gross_sales_mln
percentage

OUTPUT

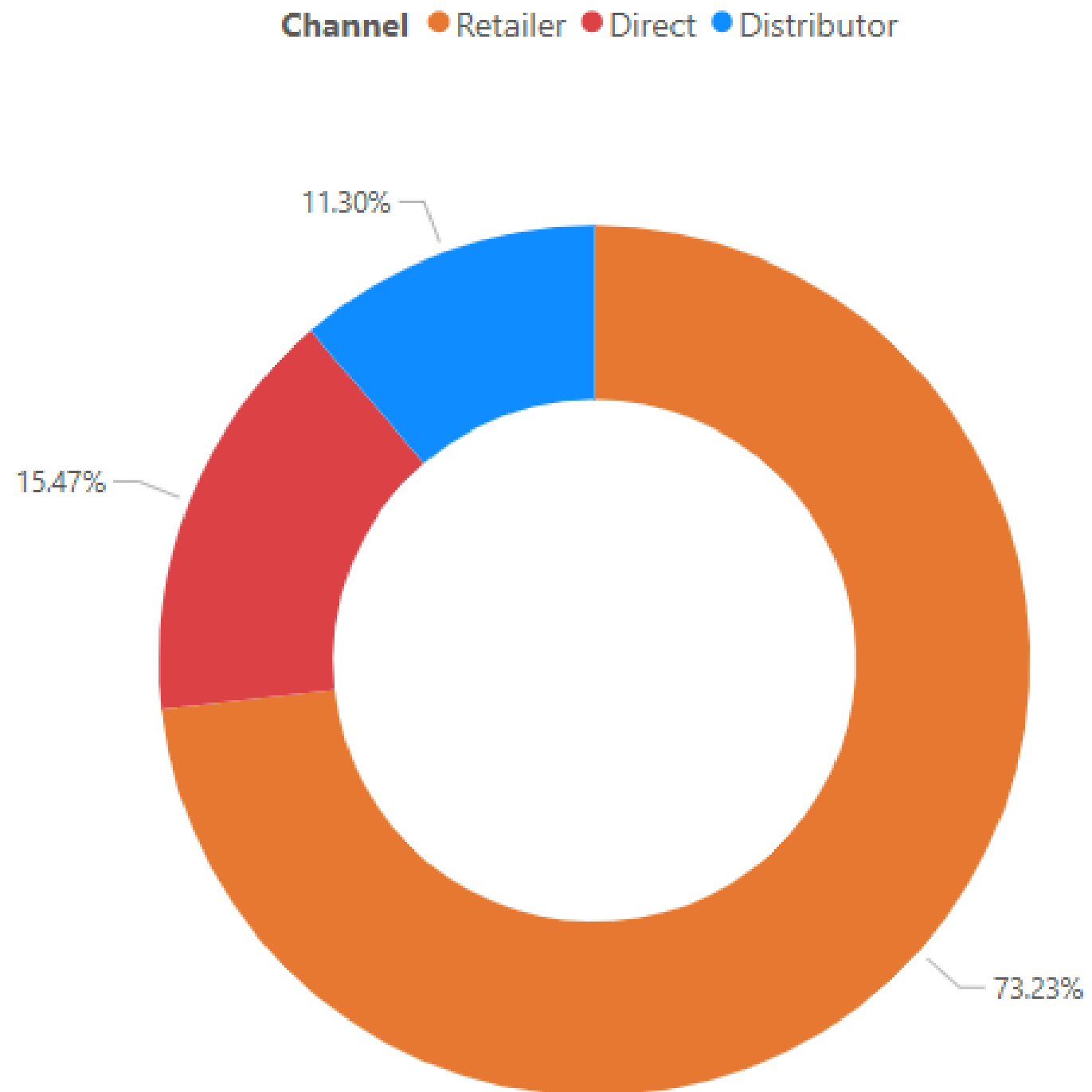
	channel_	gross_sales_million	percentage
▶	Retailer	\$1219.08	73.23%
	Direct	\$257.53	15.47%
	Distributor	\$188.03	11.30%



INSIGHTS

Retailers accounted for the majority of sales, representing 75% of the total.

Direct channels contributed only 11% of total sales.



REQUEST 10.

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

OUTPUT

The final output contains these fields:

division
product_code
Product
total_sold_quantity
rank_order

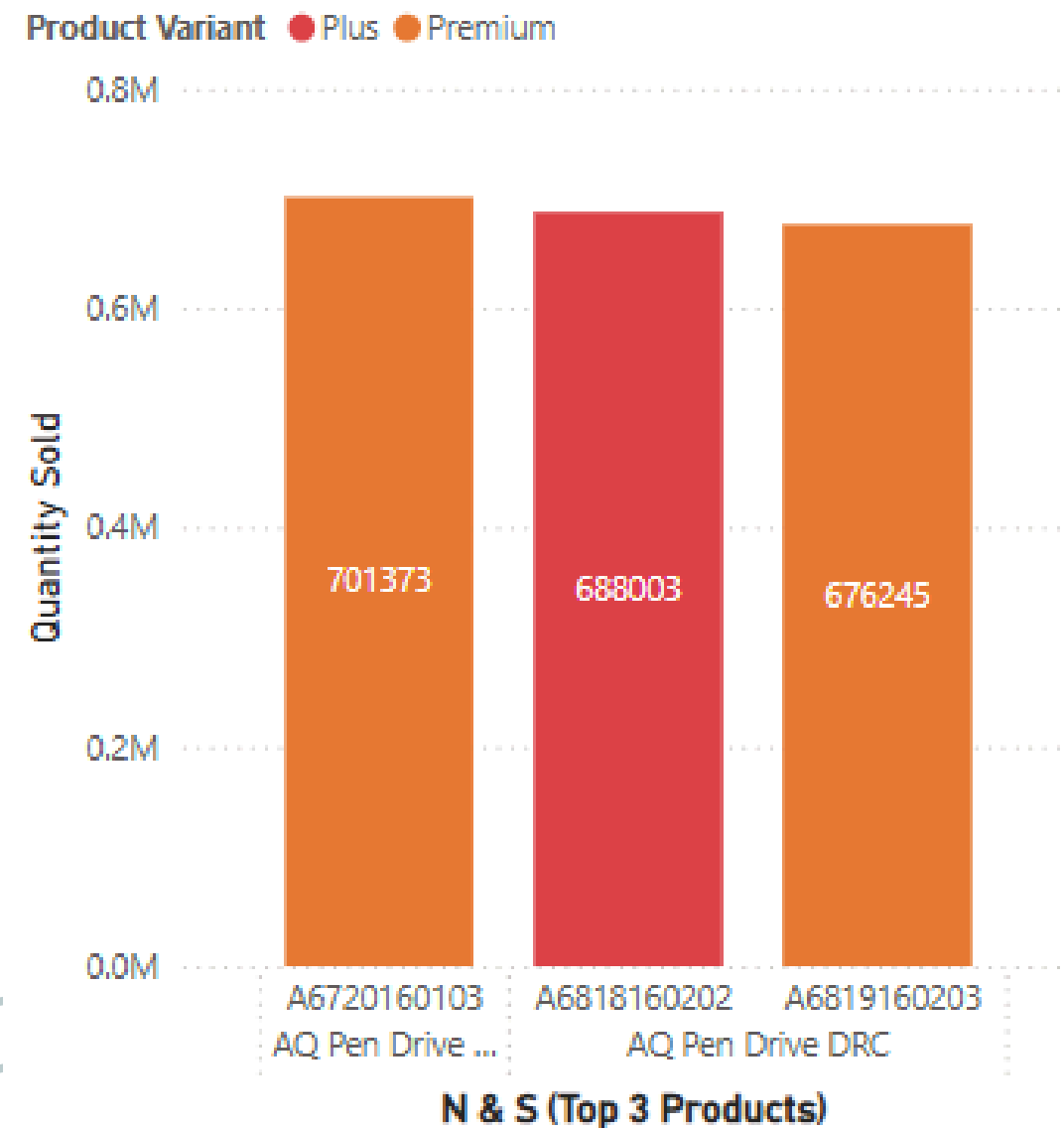
	division	product_code	product	total_sold_quantity	rank_order
▶	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
	N & S	A6818160202	AQ Pen Drive DRC	688003	2
	N & S	A6819160203	AQ Pen Drive DRC	676245	3
	P & A	A2319150302	AQ Gamers Ms	428498	1
	P & A	A2520150501	AQ Maxima Ms	419865	2
	P & A	A2520150504	AQ Maxima Ms	419471	3
	PC	A4218110202	AQ Digit	17434	1
	PC	A4319110306	AQ Velocity	17280	2
	PC	A4218110208	AQ Digit	17275	3

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INSIGHTS

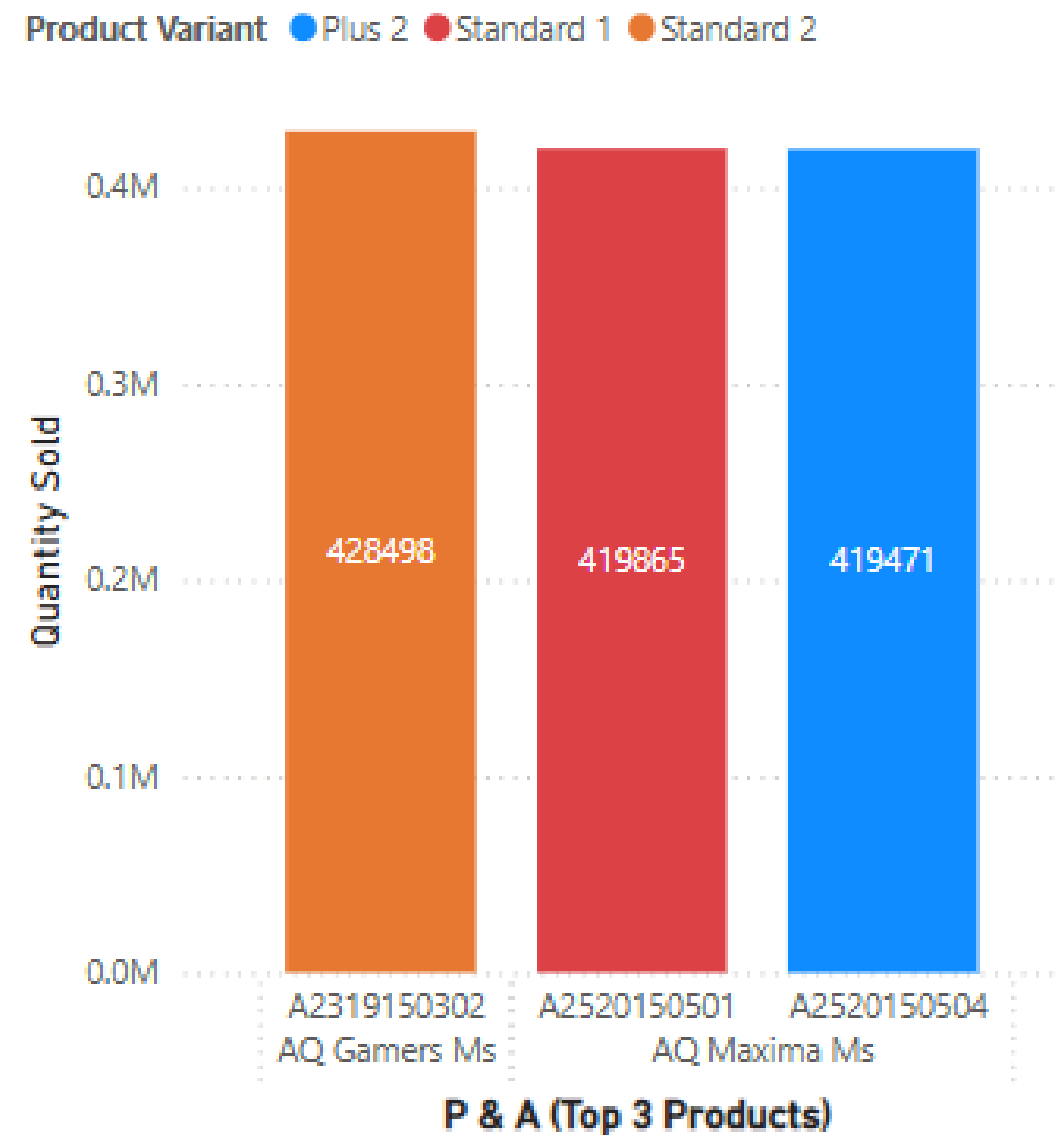
Top 3 selling products in N&S:

- AQ Pen Drive 2 IN 1
- AQ Pen Drive DRC (Plus)
- AQ Pen Drive DRC (Premium)



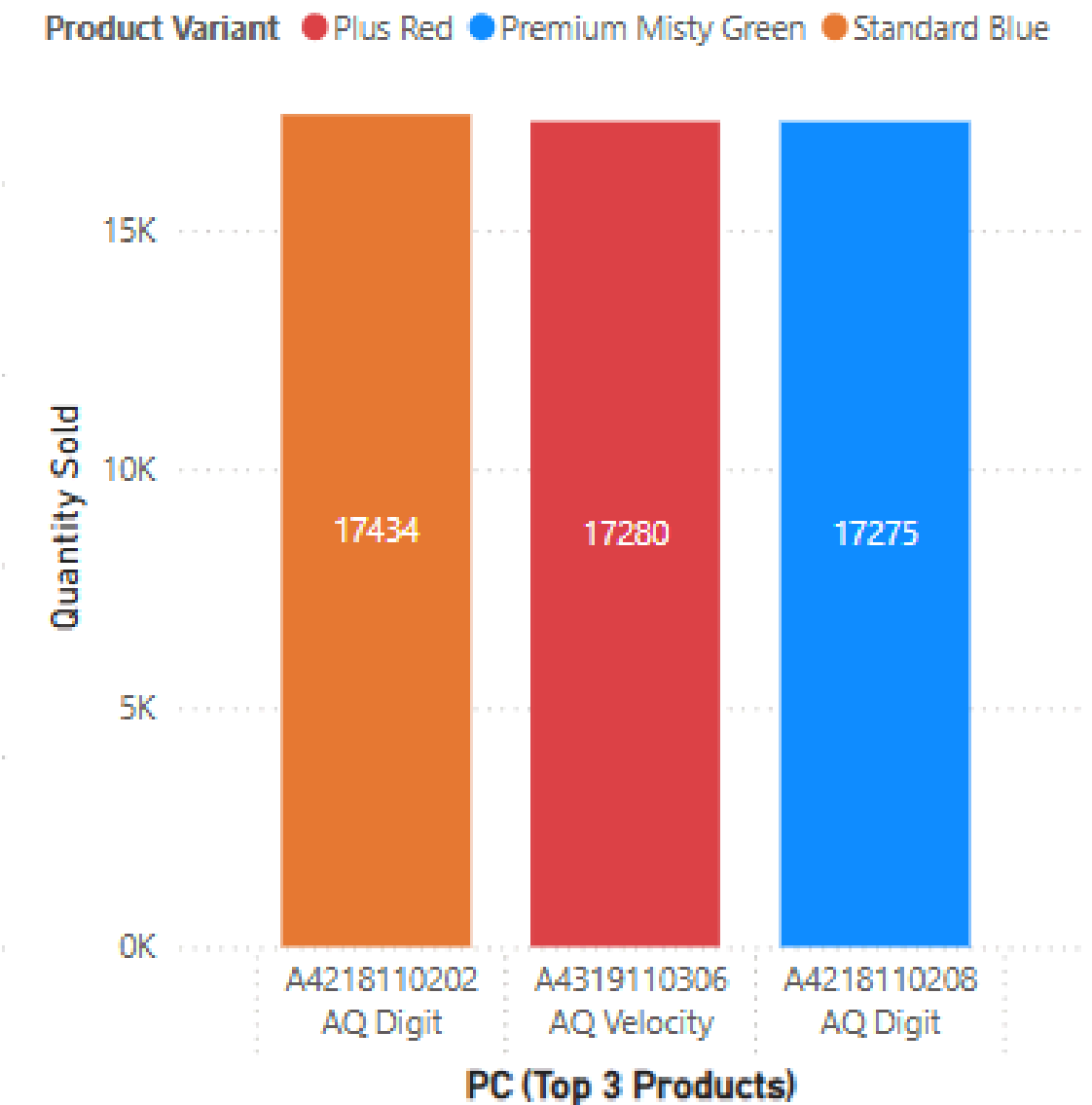
Top 3 selling products in P&A:

- AQ Gamers Ms (Standard 2)
- AQ Maxima Ms (Standard 1)
- AQ Maxima Ms (Plus 2)



Top 3 selling products in PC:

- AQ Digit (Standard Blue)
- AQ Velocity (Plus Red)
- AQ Digit (Premium Misty Green)



THANKYOU

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